

1D CNN — Architecture Variant Comparison

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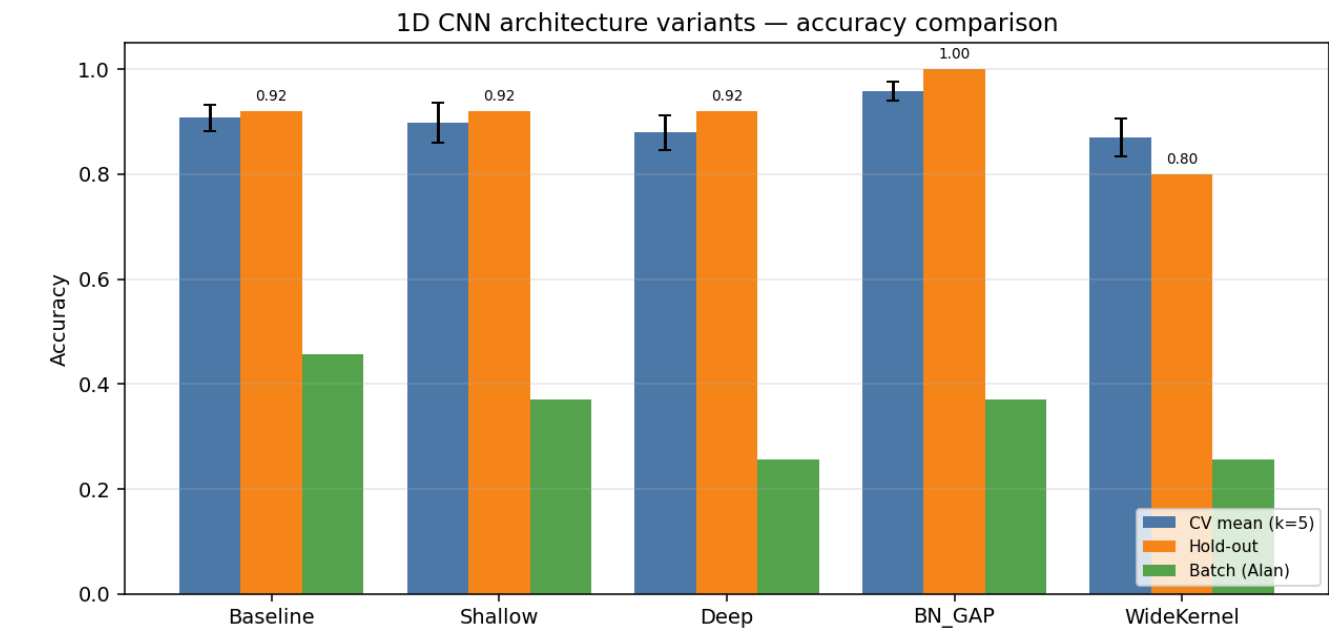
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/Anghad/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/Harry/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/Bella/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/Jestin/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/Marcus/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/Raquel/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/Tash/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/Alan/Dynamic
Total trials	241 (train pool 216, hold-out 25)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · amplitude_scale, baseline_offset
Epochs · batch size	15 · 16

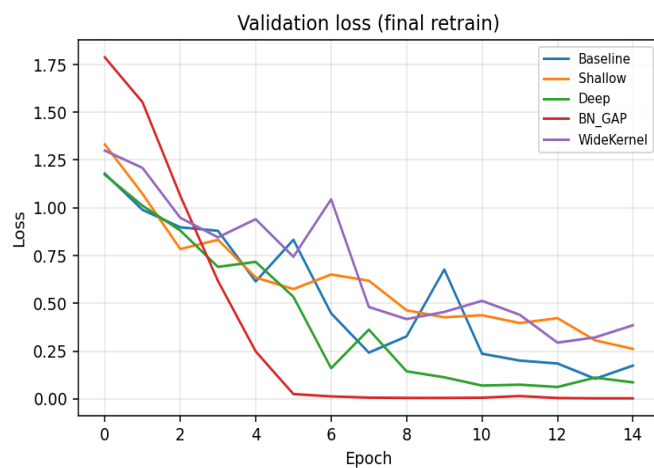
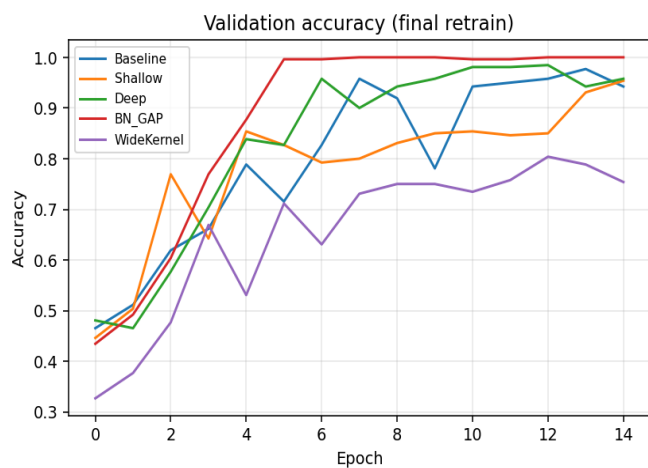
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,255	0.9074	0.0255	0.9200	0.3334	0.4571	16/35
Shallow	72,487	0.8981	0.0379	0.9200	0.4332	0.3714	13/35
Deep	204,007	0.8798	0.0333	0.9200	0.3836	0.2571	9/35
BN_GAP	57,319	0.9582	0.0175	1.0000	0.0132	0.3714	13/35
WideKernel	131,687	0.8701	0.0354	0.8000	0.3669	0.2571	9/35

Best on hold-out: **BN_GAP** · Best on CV: **BN_GAP**



Training curves (final retrain on full training pool)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
CV mean ± std	0.9074 ± 0.0255
Hold-out test acc / loss	0.9200 · 0.3334
Batch-test acc	0.4571 (16/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.9091	0.9545	0.9091
2	1038	43	0.9302	0.9302	0.9302
3	1038	43	0.9070	0.9070	0.9070
4	1038	43	0.8605	0.8605	0.8605
5	1038	43	0.9302	0.9535	0.9302

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	4
Double_Nothing	1.000	0.333	0.500	3
Double_Pistol_Recoil	0.800	1.000	0.889	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.750	1.000	0.857	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.936	0.905	0.892	25
weighted avg	0.938	0.920	0.905	25

Batch test per class (Alan)

Class	Correct	Total	Accuracy
Double_Lower	3	5	0.600
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	0	5	0.000
Double_Raise	0	5	0.000
Double_Wiggle	4	5	0.800
Double_cmere	5	5	1.000
Drum_Roll	0	5	0.000
Overall	16	35	0.457

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
CV mean ± std	0.8981 ± 0.0379
Hold-out test acc / loss	0.9200 · 0.4332
Batch-test acc	0.3714 (13/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.9091	0.9091	0.9091
2	1038	43	0.9070	0.9070	0.9070
3	1038	43	0.8837	0.9070	0.8837
4	1038	43	0.8372	0.8372	0.8372
5	1038	43	0.9535	0.9535	0.9535

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	4
Double_Nothing	1.000	0.333	0.500	3
Double_Pistol_Recoil	0.800	1.000	0.889	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.750	1.000	0.857	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.936	0.905	0.892	25
weighted avg	0.938	0.920	0.905	25

Batch test per class (Alan)

Class	Correct	Total	Accuracy
Double_Lower	0	5	0.000
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	0	5	0.000
Double_Raise	0	5	0.000
Double_Wiggle	4	5	0.800
Double_cmere	5	5	1.000
Drum_Roll	0	5	0.000
Overall	13	35	0.371

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
CV mean ± std	0.8798 ± 0.0333
Hold-out test acc / loss	0.9200 · 0.3836
Batch-test acc	0.2571 (9/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.8409	0.9773	0.8409
2	1038	43	0.9070	1.0000	0.9070
3	1038	43	0.9070	0.9070	0.9070
4	1038	43	0.8372	0.8837	0.8372
5	1038	43	0.9070	0.9070	0.9070

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	4
Double_Nothing	1.000	0.333	0.500	3
Double_Pistol_Recoil	0.800	1.000	0.889	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.750	1.000	0.857	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.936	0.905	0.892	25
weighted avg	0.938	0.920	0.905	25

Batch test per class (Alan)

Class	Correct	Total	Accuracy
Double_Lower	4	5	0.800
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	0	5	0.000
Double_Raise	0	5	0.000
Double_Wiggle	0	5	0.000
Double_cmere	1	5	0.200
Drum_Roll	0	5	0.000
Overall	9	35	0.257

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
CV mean ± std	0.9582 ± 0.0175
Hold-out test acc / loss	1.0000 · 0.0132
Batch-test acc	0.3714 (13/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.9773	1.0000	0.9773
2	1038	43	0.9767	1.0000	0.9767
3	1038	43	0.9302	0.9535	0.9302
4	1038	43	0.9535	0.9767	0.9535
5	1038	43	0.9535	0.9767	0.9535

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	4
Double_Nothing	1.000	1.000	1.000	3
Double_Pistol_Recoil	1.000	1.000	1.000	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	1.000	1.000	1.000	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	1.000	1.000	1.000	25
weighted avg	1.000	1.000	1.000	25

Batch test per class (Alan)

Class	Correct	Total	Accuracy
Double_Lower	2	5	0.400
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	0	5	0.000
Double_Raise	1	5	0.200
Double_Wiggle	5	5	1.000
Double_cmere	0	5	0.000
Drum_Roll	1	5	0.200
Overall	13	35	0.371

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
CV mean ± std	0.8701 ± 0.0354
Hold-out test acc / loss	0.8000 · 0.3669
Batch-test acc	0.2571 (9/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.9318	0.9773	0.9318
2	1038	43	0.8372	0.9302	0.8372
3	1038	43	0.8837	0.8837	0.8837
4	1038	43	0.8372	0.9070	0.8372
5	1038	43	0.8605	0.9302	0.8605

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.500	1.000	0.667	4
Double_Nothing	1.000	0.333	0.500	3
Double_Pistol_Recoil	0.800	1.000	0.889	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.000	0.000	0.000	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.757	0.762	0.722	25
weighted avg	0.768	0.800	0.749	25

Batch test per class (Alan)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	4	5	0.800
Double_Pistol_Recoil	0	5	0.000
Double_Raise	0	5	0.000
Double_Wiggle	0	5	0.000
Double_cmere	0	5	0.000
Drum_Roll	0	5	0.000
Overall	9	35	0.257